



12388 - The Vela Jr Central Compact Object: A JWST Window into Massive Star Death and Neutron Star Birth

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI MRS Background	MIRI Medium Resolution Spectroscopy	(2) Background
	2	MIRI MRS CCO Nebula	MIRI Medium Resolution Spectroscopy	(1) VelaJr-CCO-Nebula

ABSTRACT

The progenitors of the Central Compact Objects (CCOs) — an enigmatic class of neutron stars found in young supernova remnants (SNRs) — remain a subject of significant interest and debate. This growing class, studied almost exclusively in X-rays, provides key insights into neutron star formation and evolution. Despite their youth, CCOs lack the wind nebulae typically associated with young neutron stars. The Vela Jr SNR, which hosts the CCO CXOU J085201.4-461753 — the only Galactic CCO associated with a compact optical nebula — offers a unique opportunity to investigate both the progenitors of CCOs and the environment of non-thermal dominated SNRs. Recent optical integral field unit (IFU) observations reveal a nitrogen-rich environment indicative of a massive progenitor — possibly a Wolf-Rayet star — with preliminary modeling suggesting an initial mass of ~ 100 solar masses. If confirmed, this would imply that some of the most massive stars do not end their lives black holes, challenging standard core-collapse paradigms. We propose JWST MIRI MRS observations to map the Vela Jr.'s CCO nebula in the mid-infrared, focusing on the Sulfur and Neon lines that trace the nebula's structure, composition, and kinematics in the mid-infrared — thus linking progenitor mass loss, CCO properties, and the conditions that give rise to efficient particle acceleration. This JWST program will potentially establish Vela Jr as a benchmark system for linking massive star progenitors, compact object formation, and the origin of non-thermal dominated SNRs.

OBSERVING DESCRIPTION

The proposed JWST MIRI MRS observation will target the Vela Jr CCO nebula to obtain the complete mid-infrared spectral coverage from 4.9-27.9 μm , focusing on the diagnostic lines [Ne II] 12.81 μm , [Ne III] 15.56 μm , [S IV] 10.51 μm , and [S III] 18.71 μm . These extinction-free fine-structure lines are powerful tracers of the nebular ionization structure and elemental yields, providing direct constraints on the progenitor's mass. To fully cover the spatial extent of the nebula, the observation will employ a 2x2 mosaic pattern using all four MRS channels and the three grating settings.

We request a total exposure time of 6.6 hours (3.1 hours of science time). These observations will deliver the first spatially resolved mid-infrared maps of the nebula's emission structure, composition, and kinematics. In addition to the primary diagnostic ratios [Ne III]/[Ne II] and [S IV]/[S III], MIRI MRS will detect several other key transitions (e.g., [Fe II], [Ar II], [Ar III], [Ni II]), enabling a comprehensive analysis of the excitation conditions and ionization mechanisms. The resulting dataset will constrain potential expansion or motion relative to the CCO, and, when combined with MUSE optical data, provide a complete multi-wavelength picture of the system, a crucial step toward understanding CCO progenitors and their environments.

Proposal 12388 - Targets - The Vela Jr Central Compact Object: A JWST Window into Massive Star Death and Neutron Star Birth

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	VelaJr-CCO-Nebula	RA: 08 52 1.4200 (133.0059167d) Dec: -46 17 54.52 (-46.29848d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Target RA and DEC of the Background has been chosen from archival MUSE observation of Vela Jr. central area. Category=ISM Description=[Compact nebulae, Emission nebulae, Gaseous nebulae] Extended=YES				
Fixed Targets	(2)	Background	RA: 08 51 55.7991 (132.9824962d) Dec: -46 19 0.92 (-46.31692d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Target RA and DEC of the Background has been chosen from archival MUSE observation of Vela Jr. central area. Category=Unidentified Description=[Blank field] Extended=YES				

Proposal 12388 - Observation 1 - The Vela Jr Central Compact Object: A JWST Window into Massive Star Death and Neutron Star Birth

Observation	Proposal 12388, Observation 1: MIRI MRS Background												Wed Jun 24 22:00:10 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [MIRI MRS CCO Nebula (Obs 2)]												
Diagnostics	(MIRI MRS Background (Obs 1)) Warning (Form): Imager Filter overlap.												
	(MIRI MRS Background (Obs 1)) Warning (Form): Imager Filter overlap.												
	(MIRI MRS Background (Obs 1)) Warning (Form): Imager Filter overlap.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	Background	RA: 08 51 55.7991 (132.9824962d)					Epoch of Position: 2000					
			Dec: -46 19 0.92 (-46.31692d)										
			Equinox: J2000										
			Comments: Target RA and DEC of the Background has been chosen from archival MUSE observation of Vela Jr. central area.										
Acquisition	Category=Unidentified												
	Description=[Blank field]												
Template	Extended=YES												
	#	Target											
Dithers	1	NONE											
	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
Spectral Elements		All MRS				YES			FULL		Allow Auto Reorder		
	#	Dither Type					Optimized For			Direction			
Spectral Elements	1	4-Point					BACKGROUND			NEGATIVE			
	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1280W	FASTR1	5	5	1	Dither 1	4	20	321.905	
	1	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011	
	1	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011	
	2		IMAGER	F1500W	FASTR1	5	5	1	Dither 1	4	20	321.905	
	2	LONG(C)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011	
	2	LONG(C)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011	
	3		IMAGER	F1800W	FASTR1	5	5	1	Dither 1	4	20	321.905	
	3	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
3	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 12388 - Observation 1 - The Vela Jr Central Compact Object: A JWST Window into Massive Star Death and Neutron Star Birth

Special Requirements	Aperture PA Range 290.0 to 135.0 Degrees (V3 290.0 to 135.0) Sequence Observations 1, 2, Non-interruptible
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Proposal 12388 - Observation 2 - The Vela Jr Central Compact Object: A JWST Window into Massive Star Death and Neutron Star Birth

Observation	Proposal 12388, Observation 2: MIRI MRS CCO Nebula												Wed Jun 24 22:00:10 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI MRS Background (Obs 1)]													
Diagnostics	(MIRI MRS CCO Nebula (Obs 2)) Warning (Form): Imager Filter overlap.													
	(MIRI MRS CCO Nebula (Obs 2)) Warning (Form): Imager Filter overlap.													
	(MIRI MRS CCO Nebula (Obs 2)) Warning (Form): Imager Filter overlap.													
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	VelaJr-CCO-Nebula	RA: 08 52 1.4200 (133.0059167d)				Epoch of Position: 2000							
			Dec: -46 17 54.52 (-46.29848d)											
			Equinox: J2000											
	Comments: Target RA and DEC of the Background has been chosen from archival MUSE observation of Vela Jr. central area. Category=ISM Description=[Compact nebulae, Emission nebulae, Gaseous nebulae] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %			Row shift (deg)		Column shift (deg)		Tile Order	
	2	2	10.0			10.0			0.0		0.0		DEFAULT	
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F1280W	FASTR1	5	5	1	Dither 1	4	20	321.905		
	1	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011		
	1	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011		
	2		IMAGER	F1500W	FASTR1	5	5	1	Dither 1	4	20	321.905		
	2	LONG(C)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011		
	2	LONG(C)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011		
	3		IMAGER	F1800W	FASTR1	5	5	1	Dither 1	4	20	321.905		
	3	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Proposal 12388 - Observation 2 - The Vela Jr Central Compact Object: A JWST Window into Massive Star Death and Neutron Star Birth

Special Requirements	Aperture PA Range 290.0 to 135.0 Degrees (V3 290.0 to 135.0) Sequence Observations 1, 2, Non-interruptible
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